1. End-User Requirements Analysis

1.1 Decisions this end user makes

Gunasekaran et al. (2004) argue that supply-chain measures matter because they feed strategic, tactical and

operational planning and control, including setting objectives and evaluating performance. A regional operations head sits at the tactical-operational boundary: the network is given, but service levels, shipping-mode use and the order of remedial effort are decided here. Wang et al. (2016) classify supply-chain analytics as descriptive, predictive or prescriptive and by strategic versus operational focus. With historical order records only, descriptive and diagnostic evidence is both feasible and decision-relevant; prescriptive optimization would claim more than the data supports. The expected decisions are therefore: D1 is delivery performance acceptable against target and is it deteriorating; D2 where (region, mode, product) should corrective effort go first; D3 are delivery promises realistic, and which shipping services should be redesigned or re-priced; D4 is poor delivery costing margin, and so how much remediation is justified.

1.2 Evidence and measures required

Rao et al. (2011) show, for previously loyal customers of an online retailer, that a fulfilment delay reduced order frequency and order size and raised customer anxiety, interpreted through expectation disconfirmation. The implication for evidence is that the gap between the promise and the outcome matters, not average speed alone: the dashboard must show scheduled versus actual days and the size of the gap. Gunasekaran et al. (2004) stress balancing time-based delivery measures with cost and financial measures, so delivery evidence must sit beside sales, profit, margin and loss-making share. The key indicators are: late-delivery rate, promised versus actual days, days late, order volume and sales, profit and margin, and share of loss-making lines.

1.3 Level of detail and investigation

Yigitbasioglu and Velcu (2012) conclude that dashboards succeed when visualization principles manage presentation format and information load, and when the dashboard offers some flexibility. For this user the appropriate structure is layered: headline indicators first, then comparison across the dimensions that can be acted on (time, region, shipping mode, department), then drill-down. Order-level detail is unnecessary at this role, but region x shipping-mode is the finest grain that maps to a decision. Hazen et al. (2014) show that poor data quality undermines supply-chain analytics. A manager who cannot see coverage gaps or small samples will over-read noise, so the

dashboard must expose sample size, coverage warnings and uncertainty.

over-read noise, so the dashboard must expose sample size, coverage warnings and uncertainty.

1.4 Derived requirements and how they shaped the dashboard

Seven requirements followed from the literature and shaped the dashboard. **R1** monitor performance against target and overtime (Gunasekaran et al., 2004; Yigitbasioglu & Velcu, 2012), met by the target-colored KPI strip and the V1 trend with an adjustable target line. **R2** shows the promise-versus-actual gap rather than averages (Rao et al., 2011), met by the V3 diverging bars and the V2 'days late vs promise' metric. **R3** locate problems by region and service, then focus (Wang et al., 2016; Yigitbasioglu & Velcu, 2012), met by the V2 heat-map whose click cross-filters all other views. **R4** priorities by business scale (Gunasekaran et al., 2004), met by the V4 treemap with drill-down. R5 connect delivery to profit and cost leakage (Gunasekaran et al., 2004; Rao et al., 2011), met by V5 with 95% confidence intervals. **R6** makes data limits visible (Hazen et al., 2014), met by small-sample suppression, a coverage band and a Data notes dialog. R7 keeps exploration flexible and light on cognitive load (Yigitbasioglu & Velcu, 2012), met by global filters, three stages, reset and CSV export.

1.5 Critical reflection on the evidence base

The literature is used as evidence, not as a feature checklist. Rao et al. study one small printed-material retailer, so effect sizes cannot be transferred; the dataset has no customer repeat-purchase history, so the retention cost they describe cannot be measured here (a limitation carried into Section 5). Yigitbasioglu and Velcu note that little is known about which dashboard types work for which users, so each interaction is justified by an analytical purpose rather than by popularity. Gunasekaran et al. draw on UK companies in the early 2000s, before e-commerce delivery promises were common, and Wang et al. emphasize big-data scale that this 173,000-line dataset does not reach. These limits justify a descriptive-diagnostic dashboard with explicit uncertainty rather than predictive claims.

2. Problem Statement

2.1 Initial investigation of the data

Profiling the dataset (180,519 lines; 7,754 cancelled or suspected-fraud lines never shipped and were excluded, leaving 172,765 lines worth US$35.2M) showed that 57.3% of shipped lines were late (2.9 days promised versus 3.5 actual), that the rate is almost identical across markets (56.8-57.7%) and months (55.2-59.3%), and that it differs sharply

by shipping mode (First Class 100%, Second Class 80%, Same Day 48%, Standard 40%). Markets are also reported in different periods, a coverage limitation.

2.2 Problem statement

A Regional Head of E-Commerce Operations oversees fulfilment for five markets and 23 regions, which between January 2015 and January 2018 shipped 172,765 order lines worth US$35.2M across four shipping modes, each with a scheduled delivery promise (context). 57.3% of these lines arrive later than promised, and the rate barely changes with market, region, department or month; because it differs sharply by shipping mode (First Class promises 1 day but always takes 2), the problem appears to lie in how promises are set rather than in local execution, so a small number of service-design changes could remove most of the lateness (problem/opportunity). This matters because US$20.1M of sales (57.2%), about 2,900 late lines a month, travelled on late deliveries, and Rao et al. (2011) show that broken promises reduce later order frequency and size, so the hidden cost may exceed the profit visible in the data (significance). The scope is shipped lines from 2015 to 2018 analyzed by month, market/region, mode, department and delivery outcome, excluding carrier cost, customer retention and causal effects, which the data does not contain (scope). The analysis therefore aims to show the Regional Head where and how to redesign delivery promises first, by establishing whether lateness is episodic or structural (Q1), where it concentrates (Q2), why it occurs, i.e. promise versus actual time (Q3), which parts of the business carry the most sales (Q4), and whether late delivery or discounting erodes profit (Q5) (analytical purpose).

One-sentence problem statement. The Regional Head of E-Commerce Operations needs to understand why 57.3% of shipped order lines (US$20.1M of sales) arrive late when the rate is nearly identical across markets, regions and months, so that delivery promises and shipping modes can be redesigned first where the most business is exposed.

2.3 How the questions map to the five visualizations

V1 (late-delivery trend with volume) answers Q1, V2 (region x shipping-mode heat-map) answers Q2, V3 (diverging bars of actual minus scheduled days) answers Q3, V4 (market-region-department treemap) answers Q4, and V5 (profit by discount band, late versus not late, with 95% confidence intervals) answers Q5; together they serve requirements R1 to R6.

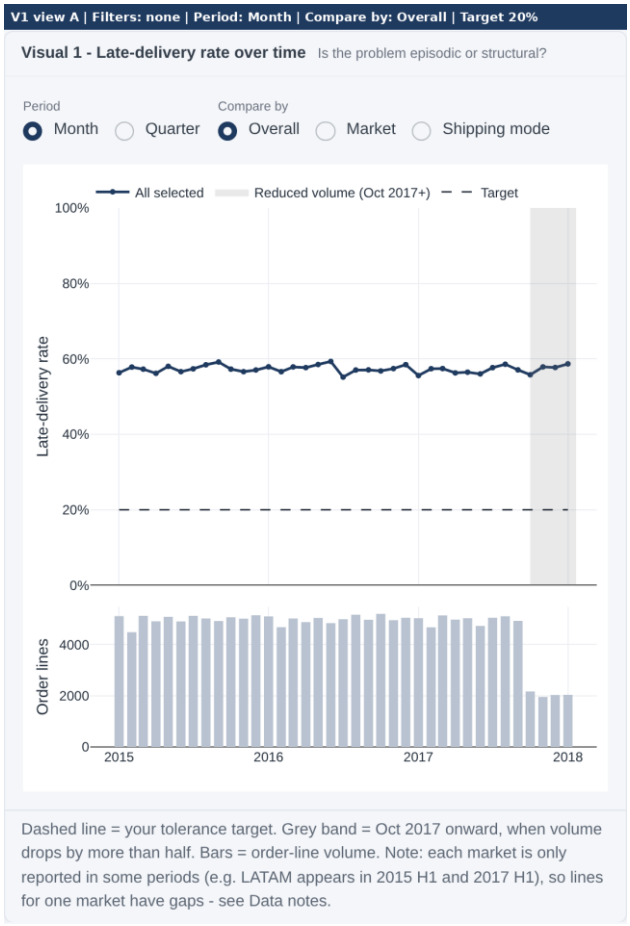
4. Five Visualization Interpretations

4.0 Dashboard architecture and interactions

The dashboard follows the decision flow of the Regional Head in three stages: Monitor and Locate (V1, V2), Diagnose and Prioritize (V3, V4) and Business Impact (V5 with auto-generated analyst notes), framed by a persistent filter panel and KPI strip. Clicking a heat-map cell sets a focus (region x shipping mode) that cross-filters V1, V3, V4, V5 and the KPIs, so evidence found in one view is investigated in the others. Eight interactions support this exploration: cascading global filters, click-to-focus, an adjustable late-delivery target, a metric switch on V2, a period and compare-by control on V1, treemap drill-down with size and color selectors, a measure switch on V5, and reset, data notes and CSV download.

4.1 Visualization 1: Late-delivery rate over time (line chart with aligned volume bars)

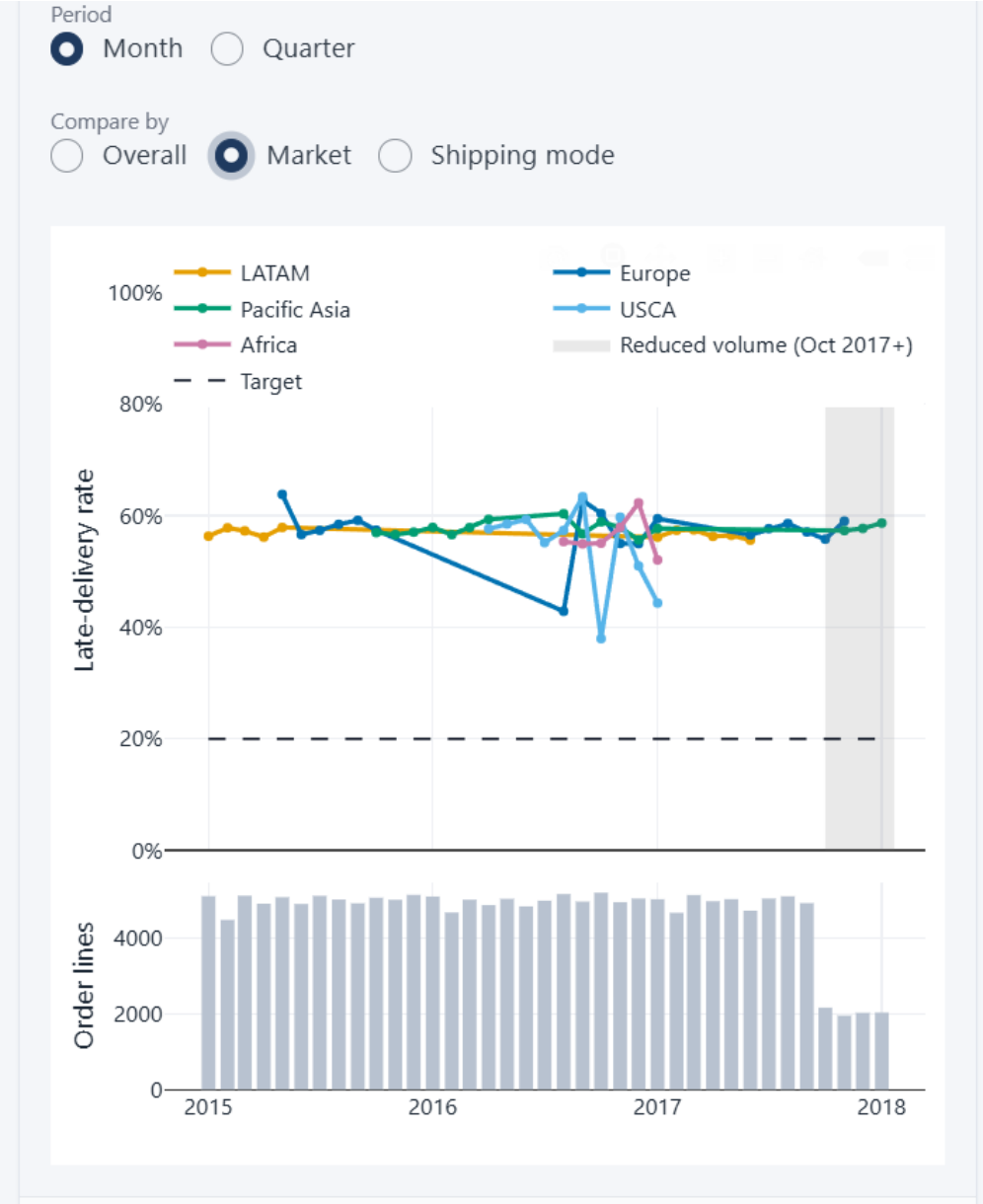
Design justification. To show whether lateness is episodic or structural, time is encoded by horizontal position and the late rate by vertical position on a common 0-100% scale. Position along a common scale is the most accurately decoded visual channel (Cleveland and McGill, 1984), and a zero baseline prevents exaggeration of small movements. Connected lines exploit the Gestalt principle of continuity, which makes trend and volatility easy to read. The dashed target line supplies context, since a KPI without a comparator is hard to interpret (Few, 2006). A volume bar panel shares the same time axis so that the reader judges each rate against its sample size, and a grey band flags the volume drop from October 2017. Color is restrained: one navy hue for the overall series, and the color-blind-safe Okabe-Ito palette only when comparing markets or modes, capped at five series to limit cognitive load. Interactions: period toggle, compare-by, global filters, focus and target.



View A - Filters: none; Month; Overall; target 20% Evidence. The overall late rate is a near-flat line between 55.2% and 59.3% across the 33 months with full coverage (mean 57.3%, SD 1.0 pp). It never approaches the 20% target. Monthly volume is steady at about 5,200 lines until September 2017 and then falls to about 2,100 (grey band). Pattern. There is no seasonal peak, no deterioration and no recovery: the highest month is June 2016 (59.3%) and the lowest July 2016 (55.2%), only 4 pp apart. Meaning. Lateness is the normal operating state, so waiting for it to fix itself is not an option and an incident-style response (chasing a bad month) would find nothing. The cause must be structural, which Q2 and Q3 test.



View B - Compare by Shipping mode Evidence. Four distinct bands that never cross: First Class sits at 100% every month; Second Class stays at about 78-85%; Standard Class is almost flat at 38-41%; Same Day is volatile (roughly 32-63%). Exception. Same Day swings because it has only about 250 lines a month, so sampling noise is large (see the volume panel); its long-run average is 48%. Meaning. The flat total in View A is the weighted sum of four stable but very different service-level outcomes. The structure is therefore not time but shipping mode, which shows lateness is a property of each mode's promise. First and Second Class, 35% of lines, generate 54% of all late lines



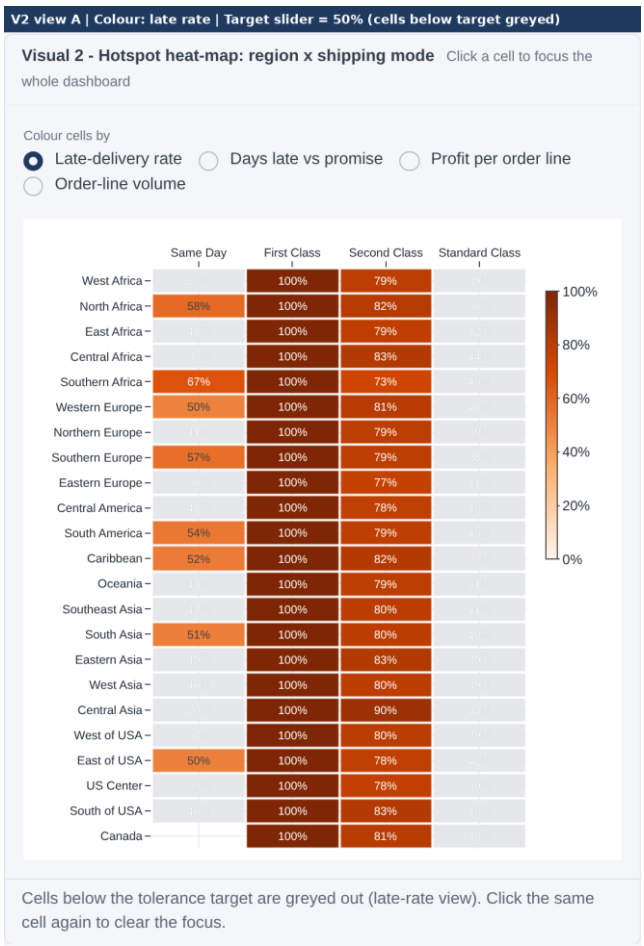
View C - Compare by Market; Quarter Evidence. Wherever a market is observed, its late rate is about 57-58%: Europe, LATAM and Pacific Asia overlap almost exactly. Exception. USCA drops to about 44% and Africa to about 52% in 2017 Q1. These points rest on very few lines (USCA has only 106 lines in 2017) and sit where those markets stop being reported, so they are coverage artefacts, not improvements. The lines also have gaps because markets are reported in different periods. Meaning. Market is not a driver of lateness, so market-by-market improvement programmed would be misdirected. Any apparent market improvement needs a sample-size and coverage check before it is believed

Comparison across views and implications. A to C shows a stable aggregate hiding stable differences by mode, with no market effect. Lateness is structural and mode-driven, which directs the user to V2 (is the mode pattern the same in every region?) and V3 (why does each mode fail?). Data limitation: market trends are confounded by uneven reporting periods, so the overall and mode series are the trustworthy trend signals.

4.2 Visualization 2: Hotspot heat-map: region x shipping mode

Design justification. To show where lateness concentrates, the 23 regions and 4 modes are placed in a matrix whose cell color encodes the measure, the most compact way to

scan two categorical dimensions for hot cells. Because color is decoded less accurately than position (Cleveland and McGill, 1984), each cell also shows its value as text. Rows are grouped by market and ordered by volume, and columns are ordered by promise speed (same day, 1, 2, 4 days) so patterns along the service ladder are visible. Late rate uses a sequential single-hue scale, “days late vs promise” uses a diverging scale centered on zero, and cells under 30 lines are left blank. The tolerance slider greys cells that meet the target so problem cells stand out. Interactions: metric switch, tolerance slider and click-to-focus.



View A - Late rate; tolerance slider set to 50% Evidence. The whole First-Class column (100% in all 23 regions) and the whole Second-Class column (73-90%) remain colored, while the entire Standard Class column (30-44%) is greyed out. Only about eight of 22 Same Day cells reach 50% (for example Southern Africa 67%, North Africa 58%, Southern Europe 57%). Pattern. Color changes down the columns far less than across them: the spread across modes (40% to 100%) dwarves the spread across regions within a mode. Meaning. Hotspots are service tiers, not places. Every region needs the same fix to First and Second Class.



View B - Days late vs promise Evidence. First Class is +1.0 days in every region; Second Class +1.9 to +2.1; Same Day +0.2 to +0.7; Standard Class between -0.3 and +0.1, i.e. delivered on or ahead of promise. Exception. Canada Standard Class is -0.3 (early); Same Day varies most (+0.2 to +0.7) because its cells are smallest. Meaning. Late-rate hotspots are not marginal slippage: Second Class misses its promise by a full two days on average, a gap large enough to damage customer expectations (Rao et al., 2011).

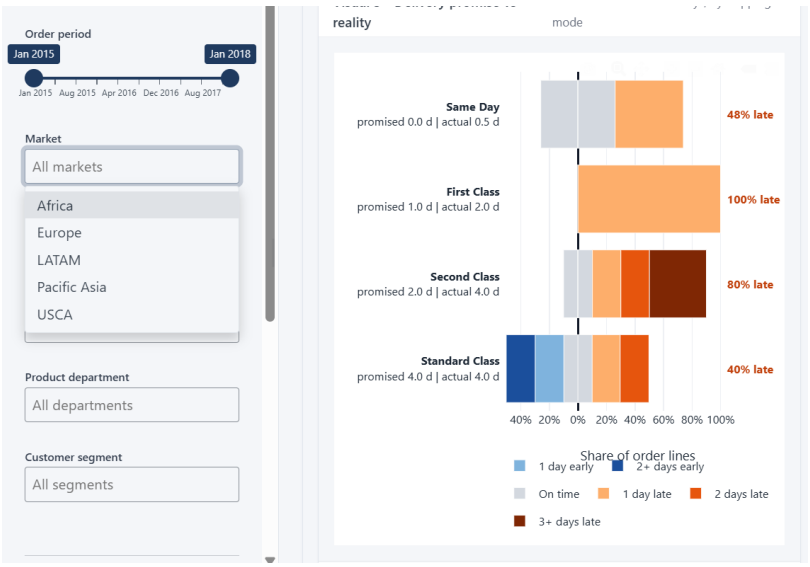


View C - Profit per order line Evidence. Cells range from $13 to $36 per line with no column pattern; mode averages are close ($21-$23). The highest cells (Canada Second Class $36, Southern Africa Second Class $35) and lowest (Eastern Asia Second Class $13) are small or noisy cells. Meaning. Profit does not follow the lateness hotspots, so hotspot status says nothing about immediate margin. The cost of lateness is therefore not captured in order-line profit; it likely sits in customer behavior (unmeasured here). Use the late rate and gap, not profit, to priorities delivery fixes.

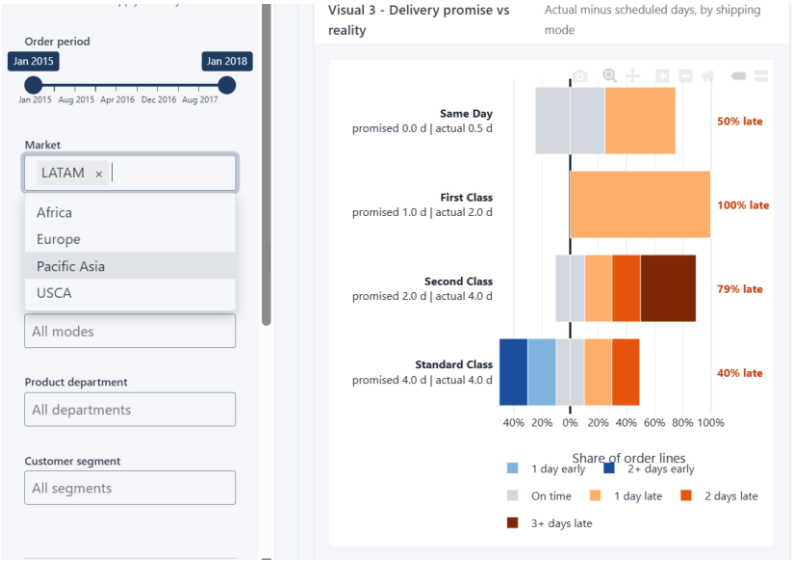
Comparison across views and implications. Across the three metrics the same message holds: mode determines the outcome in all 23 regions. Clicking a cell (for example Western Europe x Second Class) focuses the other views on that combination. The practical implication is a uniform, mode-level intervention rather than regional action plans; regional differences (51.6% Canada to 60.1% Central Africa overall) are small and partly sample noise.

4.3 Visualization 3: Delivery promise versus reality (diverging stacked bars)

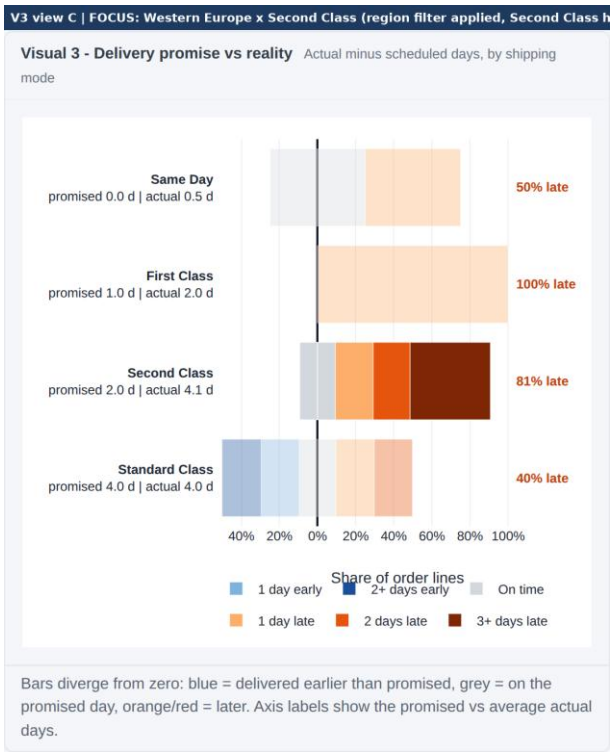
Design justification. To show whether lines are late because promises are unrealistic or delivery is slow, the chart uses the promised day as a fixed zero line: early deliveries extend left in blue and late deliveries extend right in orange to dark red. A shared baseline lets lengths on both sides be compared directly (Cleveland and McGill, 1984), the diverging scheme is color-blind safe with intensity showing severity, and grey for “on time” is deliberately low salience. Axis labels show promise versus average actual days, and each bar is annotated with its total late share, which reduces look-up effort. A pie chart (angle judgement) and four histograms (harder to compare) were rejected. Interactions: global filters and focus, which restricts a region and highlights the chosen mode.



View A - Filters: none Evidence. First Class: promised 1.0 day, actual 2.0 days, 100% exactly one day late. Second Class: promised 2.0, actual 4.0 days, 80% late, spread evenly across 0, +1, +2, +3, +4 days (about 20% each). Standard Class: promised 4.0, actual 4.0, 40% early, 20% on time, 40% late by 1-2 days. Same Day: 52% on time, 48% one day late. Pattern. Second Class and Standard Class both take 3.99 days on average with the same 2–6-day distribution, yet Second Class promises 2 days and Standard promises 4. Meaning. Second Class delivers no faster than Standard but is promised twice as fast, so its failure is a promise problem, not a delivery problem. First Class is perfectly consistent (always 2 days), so the schedule is simply wrong by one day.



View B - Filter: Market = LATAM Evidence. The shape reproduces almost exactly: 100% late First Class, 79% Second Class, 50% Same Day, 40% Standard, with the same promised and actual days. Meaning. The mechanism is not specific to one market, so corrections at the mode-policy level apply everywhere. It also strengthens confidence that View A is not an artefact of mix.

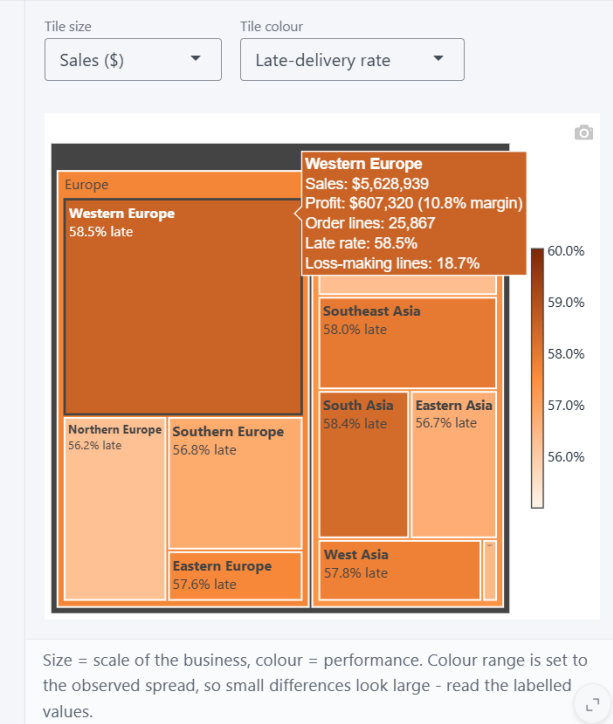


View C - Focus from V2: Western Europe x Second Class Evidence. Restricted to Western Europe, Second Class (highlighted) is 81% late with 4.1 days actual against 2.0 promised; Standard Class is 40% late; First Class 100%. Meaning. Even in the highest-sales region (16% of sales) the local picture is the network picture, so Western Europe can serve as the pilot site for a promise reset without being unrepresentative.

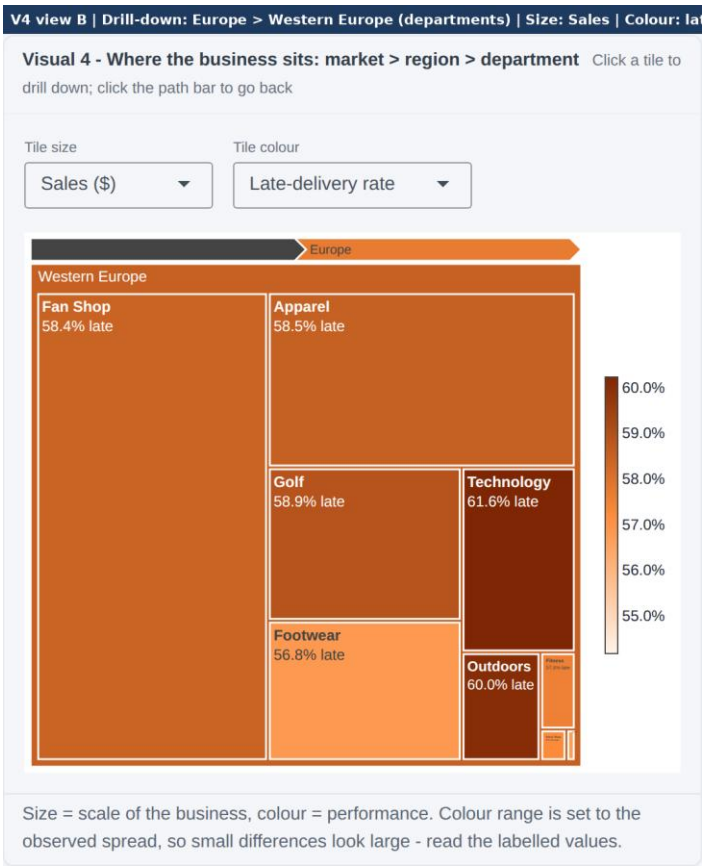
Comparison across views and implications. A mechanical what-if from the data: moving First Class's promise from 1 to 2 days would remove all of its late lines and lower the overall late rate from 57.3% to about 41.9%; if Second Class were also promised at the 4 days it typically takes, its late rate would fall from 80% to 40% and the overall rate to about 34.1%. This only re-labels the promise and does not make parcels faster, so it must be paired with a pricing and customer-impact review. It shows that most lateness is an artefact of promise design, matching the Q3 diagnosis.

4.4 Visualization 4: Where the business sits: treemap of market > region > department

Design justification. To show where sales and profit are exposed, a treemap encodes the hierarchy and part-to-whole relationship: nesting shows membership and area shows size, so the biggest businesses stand out immediately. Area is less precise than length (Cleveland and McGill, 1984), so every tile carries its label and value, and the size selector (sales, lines, profit) lets the user check rankings on another measure. Color encodes performance on a sequential single-hue scale with a labelled color bar; the range is set to the observed spread, which can exaggerate small differences, so this is noted in the card footer and values are labelled. Click-to-drill follows Shneiderman's (1996) zoom and details-on-demand and avoids showing 11 departments x 23 regions at once. Interactions: size and color selectors, drill-down, global filters, focus.



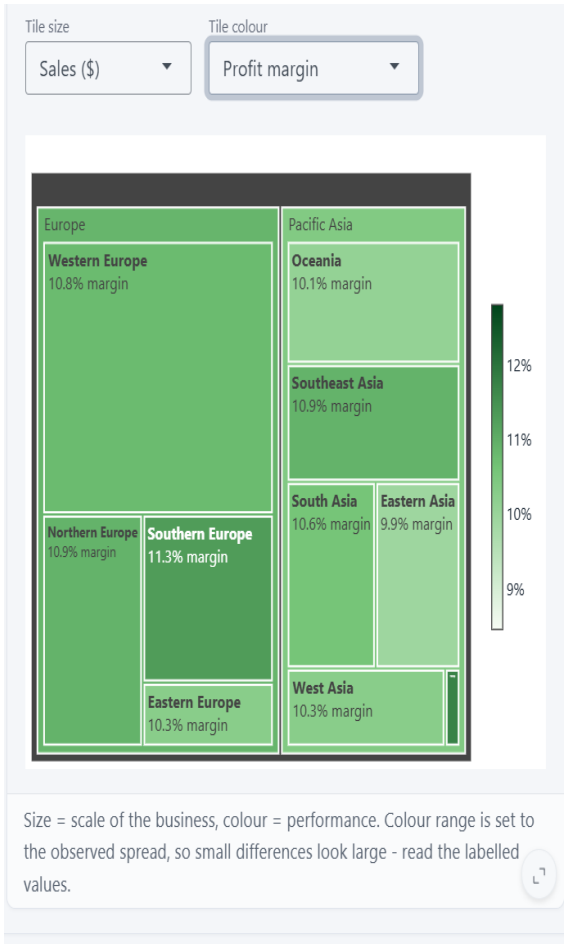
View A - Size: Sales; Color: late rate; no filters Evidence. Europe (29.5% of sales, $10.4M) and LATAM (27.9%) are the largest markets, then Pacific Asia (22.6%), USCA (13.7%) and Africa (6.3%). Western Europe (16.0%) and Central America (15.4%) alone hold 31% of sales; South America adds 8.0%. Pattern and exception. The color bar runs only 55-60%: Western Europe is darkest (58.5%), Caribbean lightest (55.5%), a 3 pp spread that looks dramatic only because of the narrow scale. Meaning. Priority should follow volume: because late rates are almost equal, the region with the most sales (Western Europe, Central America) has the latest lines and the most to gain.



View B - Drill-down: Europe > Western Europe; Size: Sales; Color: late rate Evidence. Fan Shop and Apparel are the two largest departments (late 58.4% and 58.5%). Smaller departments differ more (Technology 61.6%, Outdoors 60.0%, Footwear 56.8%) but occupy little area. Meaning. Department is not a lever either; the largest departments are late at the network average. Fixing mode promises benefits all departments, and department-specific actions would reach little volume



View C - Size: Order lines; Color: loss-making share
Evidence. The loss-making share runs from about 18.2% to 19.2% across regions (color bar 16.7-20%), with the darkest tiles small ones (for example Southern Europe departments).
Meaning. About one line in five loses money everywhere (18.7% overall), so loss-making is a pricing and discount matter, not a regional one (tested in V5).

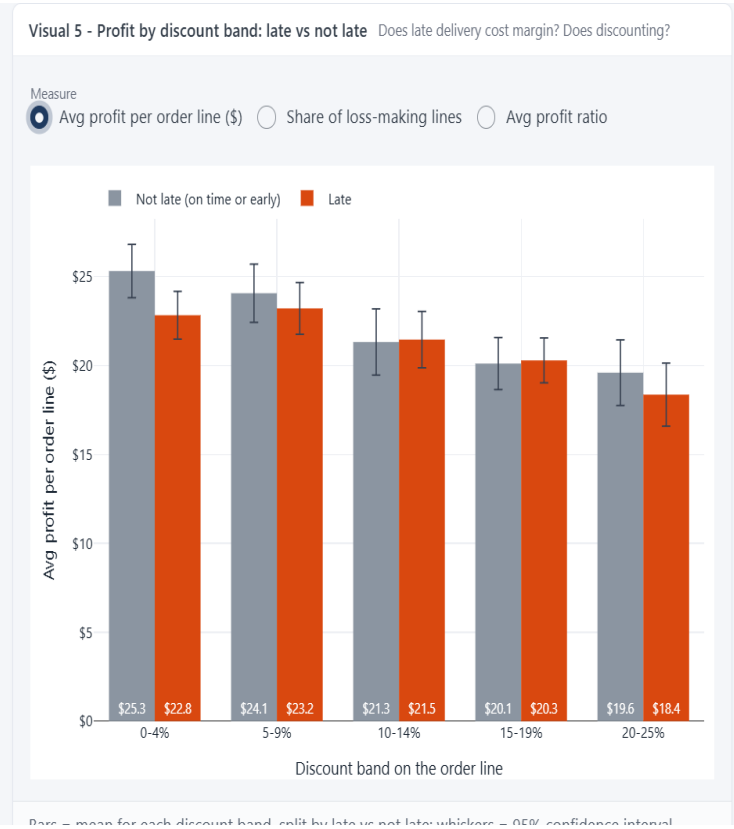


View D - Size: Sales; Color: profit margin Evidence. Margins are 10-11% across regions (color bar 9-13%): Western Europe 10.8%, Northern Europe 10.9%, South America 11.2%, Oceania 10.1%, Eastern Asia 9.9%.
Meaning. Large regions earn the network-average margin (10.8%), so delivery investment can be justified by scale rather than by an unusually profitable region.

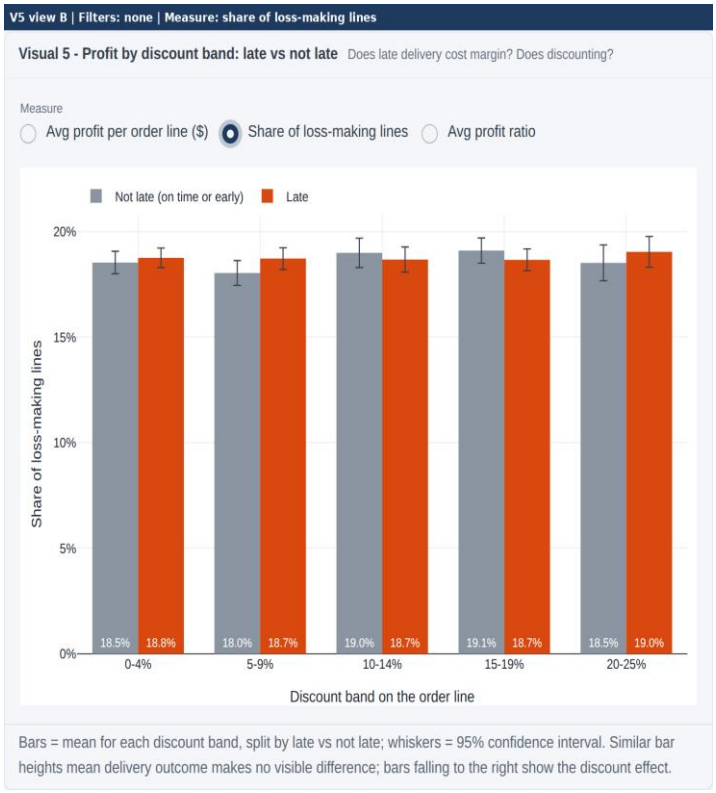
Comparison across views and implications. Across four views the pattern is consistent: performance varies by only 1-3 pp across regions, but scale varies 30-fold (Western Europe 16.0% of sales versus Central Asia 0.3%). Prioritization therefore comes from sales concentration. The Regional Head should pilot in Western Europe and Central America (31% of sales) and use the color selectors to re-check priority under lines or profit.

3.5 Visualization 5: Profit by discount band: late versus not late (grouped bars with confidence intervals)

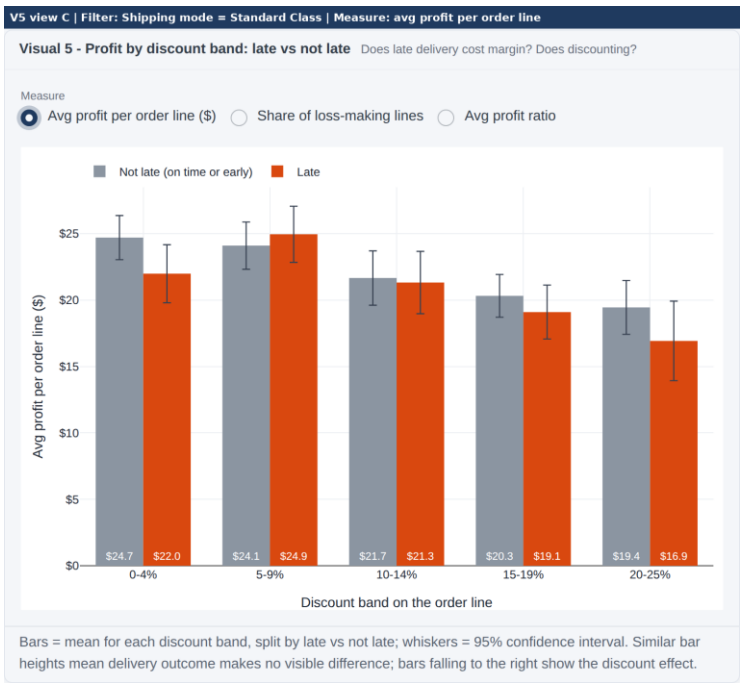
Design justification. To show whether late delivery or discounting erodes profit, mean profit is compared across five discount bands for two groups, late and not late (on-time and early combined), instead of 18 discount levels and three outcomes, which would be cluttered. Grouped bars on a zero baseline allow accurate length comparison (Cleveland and McGill, 1984), and placing the two groups side by side makes the comparison immediate. Late keeps the orange used in V3 and not late is neutral grey. 95% confidence intervals show whether a gap exceeds noise (Tufte, 2001), since profit per line has a standard deviation near $105, and values sit at the base of each bar, so they do not clash with the whiskers. A measure switch (profit, loss share, profit ratio) replaces a dual-axis chart. A 172,000-point scatter (overplotting) and boxplots (medians swamped by the large spread) were rejected.



View A - Filters: none; avg profit per order line Evidence. Profits per line fall as discount rises, from $25.3 (not late) and $22.8 (late) at 0-4% discount to $19.6 and $18.4 at 20-25%. Late and non-late bars are within about $1 of each other in the 5-9%, 10-14%, 15-19% and 20-25% bands, with overlapping intervals. Exception. The 0-4% band shows the clearest gap ($2.5), where the intervals nearly touch. Pooled across all bands, late lines earn $21.62 and not-late lines $22.58 ($p = 0.057$, Welch t-test), which is not significant at 5%. Meaning. The fitted slope is about -$0.26 per discount percentage point ($p < 0.001$), so each extra 10 pp of discount costs about $2.6 per line, whereas lateness shows no reliable penalty in recorded profit.



View B - Share of loss-making lines Evidence. The loss-making share is flat at 18-19% in every band for both groups (18.7% overall; 18.2% to 18.9% by mode); late and non-late bars are visually identical. Meaning. Roughly one line in five loses money regardless of delivery or discount level, so the loss-making population is driven by something not in the chart (likely product cost and pricing) and needs product-level review.



View C - Filter: Shipping mode = Standard Class Evidence. Holding the mode fixed, late lines are lower in the 0-4% ($22.0 vs $24.7), 15-19% ($19.1 vs $20.3) and 20-25% ($16.9 vs $19.4) bands, but higher in the 5-9% band ($24.9 vs $24.1). Every difference sits within the confidence intervals. Meaning. The direction is not consistent, so there is no reliable evidence that lateness costs profit on the current order. The expected cost is in repeat purchase (Rao et al., 2011), which needs customer-level data.

Comparison across views and implications. Profit evidence says do not justify delivery fixes by margin on the current order, and the discount tier is the visible profit lever. The case for fixing delivery must rest on customer-behavior evidence not in this dataset, a

gap addressed in Section 5. Limitation: 'Order Profit Per Order' excludes shipping cost and customer lifetime effects.

5. Data Storytelling

The story follows the INFLUENCE framework taught in the unit (Intent alignment, Narrative arc, Framing, Logic chain, Urgency, Emotional resonance, Next action, Credibility). Every claim cites the dashboard visualization that shows it.

Intent alignment: speaking to an operations head

The audience is the Regional Head of E-Commerce Operations, whose concerns are operations and performance (the COO-type lens in the framework). The story therefore leads with service performance and actions, not

strategy or finance and keeps profit as a supporting constraint.

Act I - Setup: what is the problem? (V1)

Over 33 full months, 57.3% of shipped lines arrived late. The monthly rate never left a 55-59% band (V1 view A), and no market or period stands out (V1 view C). The anchor figure is stark: US$20.1M of the US$35.2M of sales, or 98,977 order lines, travelled on a late delivery. The natural question for an operation's head is, 'Which region or team is failing?'.

Act II - Conflict: what does the data reveal? (V2, V3, V4)

Regions and markets show little difference in late-delivery rates (V4, View A). The heat map (V2) shows shipping mode is the main factor: First Class is 100% late, Second Class 73%–90%, and Standard Class 30%–44%. The diverging bars (V3) explain why: First Class takes two days despite promising one, while Second Class takes four days despite promising two, like Standard Class. These unrealistic promises cause lateness across all regions. First and Second Class make up 35% of lines but 54% of late lines, so changing regional operations will not solve the problem.

Late lines earn $21.62 per line versus $22.58 for on-time lines ($p = 0.057$), while 18.7% of all lines lose money regardless of delivery outcome (V5). Discounts have a clearer cost, at about $0.26 per line per percentage point. However, Rao et al. (2011) show that broken promises can reduce future orders, a cost not captured here. Around 2,900 lines experience this problem each month.

Act III - Resolution: what should be done? (V3, V4, V5)

Action 1 - reset the First Class scheduled promise from 1 to 2 days (actual is always 2). Evidence: V3 view A and V2 view B (+1.0 days in all regions); the what-if lowers the overall late rate from 57.3% to about 41.9%.

Owner: Head of Operations with Pricing, within one month.

Action 2 - fix or re-price Second Class. Either meet the 2-day promise through carrier SLAs or promise 4 days, and do not charge a premium for a service no faster than Standard. Evidence: V3 views A-C (4.0 versus 2.0 days); together with action 1 the overall rate falls to about 34.1%. Owner: Logistics and carrier management, within one quarter.

Action 3 - pilot in Western Europe and Central America first (31% of sales); the pilot is representative because the pattern is identical everywhere. Evidence: V4 view A and V3 view C. Owner: Regional Head, starting next quarter.

Action 4 - review discount tiers above 15%, where profit per line is lowest (V5 view A: about $22.8-$25.3 at 0-4% discount versus $18.4-$19.6 at 20-25%). Owner: Commercial team, next planning cycle.

Action 5 - add customer repeat-purchase, carrier cost and order-promise data to test whether lateness costs retention and track the late rate against the target slider monthly. Evidence: V5 view C and Section 5.

Owner: Analytics, within two quarters.

Urgency, emotional resonance and credibility

Urgency. V1 shows no sign that late deliveries will improve on their own. Each month adds about 2,900 late lines, while adjusting delivery promises is cheaper and faster than changing carriers.

Emotional resonance. Each late line represents a customer promised one-day delivery but waiting two. This risks customer trust (Rao et al., 2011), rather than blaming regional teams.

Credibility and limits. The analysis excludes 7,754 canceled lines. Different market reporting periods limit trend comparisons, shipping costs are excluded from profit, and what-if estimates do not prove customer responses or causation. Fan Shop and Apparel account for 68% of sales but have similar late rates to other departments.

Next action. Approve the First Class promise reset and review the Second Class SLA this quarter, starting with a Western Europe pilot.

6. Business Analyst Transfer Challenge: Target's Stores-as-Hubs

1. Case evidence: the central business problem

Target aims to fulfil online orders faster and cheaper without harming the in-store experience. The main challenge is deciding which building should fulfil each order. Target adjusts fulfilment by location: stores with larger backrooms handle more shipping, while busy stores reduce it. In Chicago, Target concentrated shipping in six stores, stopped shipping at 18 others, and increased fulfilment-center volume. It also uses real-time inventory and staffing data to assign orders.

2. Connection to my analysis

Service design matters more than geography. My dashboard shows late rates of 40%–100% across shipping modes, compared with 52%–60% across regions (V2, V3). Similarly, Target highlights the importance of each building's role. **Promises versus outcomes.** V3 shows the gap between promised and actual delivery. Target's claim of offering next-day delivery for five times more local demand also requires checking promise reliability. **Concentration.** V4 shows two regions account for 31% of sales, like Target concentrating shipping volume in six Chicago stores. This helps priorities resources and monitoring. **Profit evidence.** V5 shows lateness has little effect on recorded profit. Therefore, Target's cost-per-unit and sales-uplift claims must be measured rather than assumed.

3. Reproduction with Target's transaction-level data

I would apply the same analytical approach, rather than copy the charts: (i) define each fulfilled order line by its fulfilment node, promised date and delivery date; (ii) compare promise gaps by node type and service (V3), as customers care about broken promises (Rao et al., 2011); (iii) use a node-by-service heat map (V2) to compare Chicago pilot stores with other metros; (iv) track weekly delivery performance and volume around the May pilot (V1); (v) use a treemap (V4) to identify major nodes by volume and cost; and (vi) compare cost per unit and profit margin by delivery outcome (V5). Throughout, I would check sample sizes, data coverage and confidence intervals, as uneven coverage can make small markets appear to improve (V1, View C).

4. Additional evidence needed that is absent from my dataset

(a) Fulfilment node identity and type, needed to attribute outcomes to each building role. (b) Cost per unit by component (labor, carrier, last mile), since 'cheaper' is the claim and my data has profit only. (c) Inventory availability and staffing levels, which drive assignment. (d) In-store sales, traffic and order-pickup performance in the stores

that stopped shipping, to test the 'not at the expense' constraint. (e) The delivery promise shown at checkout, order cut-off time, conversion and basket size, to link promise speed to sales. (f) Data for comparable metros that did not change, and pre-pilot history. (g) Customer repeat purchase, the retention cost missing from my profit measure.

5. Challenging the claim that faster delivery caused sales gains

Target reported immediate sales gains in some categories after offering faster delivery promises, but a before-and-after comparison cannot prove causation. I would use a difference-in-differences analysis comparing Chicago with matched control cities, check pre-pilot trends, and test whether larger improvements in delivery speed lead to greater sales gains. Other possible explanations include: (1) seasonal demand changes, as the pilot began in May; (2) changes in promotions, prices or product ranges; (3) Chicago's unique characteristics, which may affect results; (4) wider next-day delivery expansion elsewhere; and (5) sales shifting from store pickup or other channels rather than increasing overall sales.

6. Decision: expand or reconsider

Expand if: Difference-in-differences results show positive net sales and margin gains, with confidence intervals excluding zero, in Chicago and at least one market with different density. Cost per unit must fall after including all node costs, delivery reliability must not decline, stores that stop shipping must maintain in-store sales and pickup performance, and gains must last beyond the first few months. **Reconsider if:** Gains occur only in certain categories or seasonal periods, cost savings disappear in less dense areas, late deliveries or stock-outs increase as volume concentrates, or sales simply shift from other channels. As V3 shows, broken delivery promises can damage customer trust even when recorded profit is unaffected.

7. Transferability limitation

My dataset is a 2015-2018 multinational shipping-mode dataset in which lateness is nearly uniform across places and is set by service-level promises, with unevenly reported markets and no node-level fields. Target's problem is a dense, domestic network where lateness and cost depend on proximity, inventory and store staffing, which my data cannot show. The finding that geography does not matter might not hold for Target, and the cost-promise trade-off is different. The two businesses cannot be assumed to behave the same way, so results must be re-estimated on Target data rather than carried over.

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8. AI Acknowledgement

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- Help me explain the shipping mode.